Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 5

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

Q1. Remove Even Numbers from a Set

Write a program to create a set of integers and remove all the even numbers from the set. Display the updated set.

Q2. Check Element in Tuple

Create a tuple containing 6 animal names. Accept an animal name from the user and check whether it is present in the tuple.

Q3. Tuple Slicing

Write a Python program to accept 8 numbers from the user and store them in a tuple. Using tuple slicing, display:

1. The first four elements
2. The last three elements
3. The tuple in reverse order

Q4. Tuple Concatenation

Write a Python program to accept 3 electronic device names from the user for the first tuple and 3 different electronic device names for the second tuple. Combine the two tuples using the + operator and display the first tuple, second tuple, and the resulting combined tuple.

Q5. Common Items in Two Shopping Lists

Write a Python program to accept 5 item names from the user for the first shopping list and 5 item names for the second shopping list. Convert both lists into sets and display:

1. Items that are common to both shopping lists.
2. Items present only in the first shopping list.
3. Items present only in the second shopping list.

Q6. Tuple Element Frequency

Write a program to create a tuple of integers and display the frequency of each element present in the tuple.

Q7. Programming Language Search Using Tuple

Write a Python program to accept the names of 6 programming languages from the user and store them in a tuple. Then accept another programming language from the user and search for it in the tuple using the `index()` method. If the language is present, display its position; otherwise, display an appropriate message.

Q8. Unique Characters in a Sentence

Accept a sentence from the user and create a set containing all the **unique characters** in the sentence, excluding spaces. Display the set and the total number of unique characters.

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